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the limit of calibration because it makes it possible to extrapolate the numeric parameters from the JVP that can be used for the clinical evaluation inter and intra-patient. This is a methodology-note that describes the innovative methods developed by the author to obtain quantitative parameters obtained by the JVP with ultrasound technique. The results reported here are only illustrative and are not produced by a dedicated experiment.

1 METHODOLOGY

PRESSENTATION ECG analysis and P, R e T events detection

Over the past 30 years, several algorithms for the automatic detection of structures P, QRS and T in ECG, have been developed[Kohler(2002)]. However, these can also be analysed manually. In this work, the video clip of the IJV was analyzed frame by frame to find the sonograms in which the cursor of the ECG was at one of the events P, R or T (see Fig. 1). The time instants relating to each event were denoted t_{Pi} , t_{Ri} and t_{Ti} , where i indicates the i -th cardiac cycle between those traces. JVP quantitative evaluation The waves a, c, x, v and y and the intervals a and v are the parameters that characterize the JVP (see Fig. 2). In this paper, these parameters are denoted in terms of the values assumed by the CSA during the cardiac cycle. The values of the CSA of the IJV in correspondence of the waves a, c, x, v and y, during the i -th cycle are indicated by the parameters a_i , c_i , x_i , v_i and y_i . These parameters